CHAPTER 10

Vectors, lines and planes in space

S3, S6, S12, S15; summary S17

Before you begin · Vectors and reading space

Without a calculator, in six minutes. Answer briefly. Use Chasles' relation, coordinates and collinearity from Première. Then choose the targeted review indicated under the questions.

1. Complete $\overrightarrow{AB} + \overrightarrow{BC} = \ldots$.

2. Calculate $\overrightarrow{AB}$ if $A(1;2)$ and $B(4;-1)$.

3. Are the vectors $(2;3)$ and $(-4;-6)$ collinear?

4. Give the coordinates of the midpoint of $[AB]$.

5. Solve 2 $x - 3 = 7$.

6. In a cube ABCDEFGH , E,F,G,H above A,B,C,D, name an edge parallel to (AB).

Answers and remediation

$\overrightarrow{AC}$; $(3;-3)$; yes; $(2{,}5;0{,}5)$; $x = 5$. For example [DC], [EF] or [HG].

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Follow the path from A to B then from B to C; R04. **2.** Subtract the starting coordinates from the end coordinates; R05. **3.** Look for the same coefficient in both coordinates; R06. **4.** Take the means of both coordinates ; R05. **5.** Add 3 then divide by 2, and check by substitution. **6.** Construct and name the faces of the cube; avoid drawing conclusions from the perspective drawing alone. The R questions serve as support for review with the teacher, even if their whole-class routine is scheduled later.

Objectives

Calculate with vectors; use collinearity and coplanarity; characterise lines and planes; work in a coordinate system.

Course notes

Vector calculation

Chasles' relation $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$, vector addition and multiplication by a real number extend to space. Two non-zero vectors are collinear if one is a multiple of the other. The zero vector is collinear with every vector. Three vectors are coplanar if they belong to the same vector plane. If $\vec{u}, \vec{v}$ are not collinear, $\vec{u}, \vec{v}, \vec{w}$ are coplanar if and only if $\vec{w} = \alpha\vec{u} + \beta\vec{v}$ for some reals α, β.

Line and plane

The line through A with non-zero direction vector $\vec{u}$ is the set of points M such that $\overrightarrow{AM} = t\vec{u}$ for a real number t. The plane through A and spanned by two non-collinear vectors $\vec{u}, \vec{v}$ is the set of points M such that $\overrightarrow{AM} = s\vec{u} + t\vec{v}$ for real numbers s, t.

Three non-collinear points define a plane. Four points A, B, C, D, with A, B, C non-collinear, are coplanar if and only if $\overrightarrow{AD}$ is a linear combination of $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

Relative positions

Two lines are coincident, strictly parallel, intersecting or non-coplanar. Vectors collinear direction vectors characterise parallel lines in the broad sense. Non-collinear direction vectors do not guarantee intersection. A line and a plane: line contained in the plane, parallel and disjoint, or intersecting at a single point. Two planes: coincident, strictly parallel, or intersecting along a line.

Example in a cube. In ABCDEFGH , the faces $ABCD$ and $EFGH$ are parallel and the edges AE, BF, CG, DH correspond. The lines (AB) and (CG) are neither parallel nor intersecting: they are not coplanar.

Basis and coordinates

A basis of space is a family of three non-coplanar vectors. Every vector can be written uniquely $x\vec{\imath} + y\vec{\jmath} + z\vec{k}$. A coordinate system consists of an origin O and a basis.

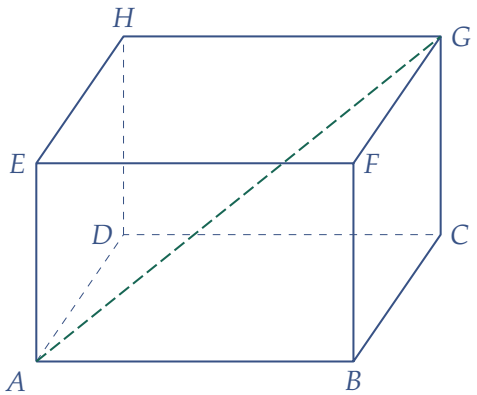
$$\overrightarrow{AB} = (x_B - x_A, y_B - y_A, z_B - z_A), \qquad I = \left(\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2}, \frac{z_A + z_B}{2}\right).$$

In an orthonormal coordinate system, $\|\vec{u}\| = \sqrt{x^2 + y^2 + z^2}$ & $AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$.

Example. In an orthonormal coordinate system, for $A(1,0,2)$, $B(3,-1,4)$, $\overrightarrow{AB} = (2,-1,2)$, $AB = 3$ and the midpoint is $(2, -\frac{1}{2}, 3)$.

Test a linear combination

Write the three coordinate equalities. Determine the coefficients from two equations when possible, then check the third. A contradiction proves non-membership; we do not delete the awkward equation.



$[AG]$ is a space diagonal, drawn in green dotted lines. The hidden edges are dotted blue; D and F are not on (AG).

Build a reliable representation

10.1 Combinations and arbitrary bases

First use: S3 and S6.

Definition · Decomposing a vector

A basis of space is a triple of non-coplanar vectors. Every vector can be written uniquely as a combination of these three vectors. A basis is not necessarily orthogonal or made up of unit vectors.

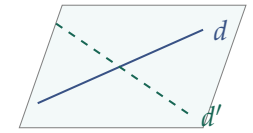
Worked example · In a tetrahedron

In a non-degenerate tetrahedron $OABC$, let $\vec{u} = \overrightarrow{OA}$, $\vec{v} = \overrightarrow{OB}$, $\vec{w} = \overrightarrow{OC}$. If I is the midpoint of $[AB]$, then $\overrightarrow{OI} = \frac{1}{2}\vec{u} + \frac{1}{2}\vec{v}$. Hence $\overrightarrow{CI} = \frac{1}{2}\vec{u} + \frac{1}{2}\vec{v} - \vec{w}$. These equalities use Chasles' relation and the midpoint, not lengths or angles.

A1 · Immediate application With the same basis, J is the midpoint of $[OC]$ and K satisfies $\overrightarrow{OK} = 2\vec{u} - \vec{v}$. Express $\overrightarrow{JK}$ then $\overrightarrow{IK}$.

10.2 Seeing is not enough to conclude

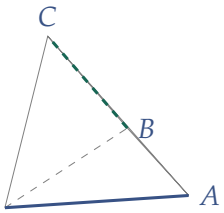
First use: S12.



intersecting, same plane



parallel, same plane



non-coplanar edges

Property · Relative positions of two lines

Two lines are either coplanar (intersecting or parallel) or non-coplanar. Vectors non-collinear direction vectors rule out parallelism, but do not prove an intersection.

Worked example · Two non-coplanar lines

In the tetrahedron $OABC$, the lines (OA) and (BC) are not coplanar. If a plane contained both, it would contain O, A, B, C, which would contradict the assumption that the tetrahedron is non-degenerate. An apparent intersection of their lines on the page would only be an effect of projection.

A2 · Read and argue On the three diagrams, name the relative position of the lines. Which geometric information accompanies the drawing to justify this answer? Give an example of lines that are not coplanar in a cube ABCDEFGH .

A3 · Refute "Two lines whose direction vectors are not collinear intersect." Disprove using the tetrahedron, then add a hypothesis that makes the statement true.

10.3 Characterising a plane by two directions

First use: S12 and S15.

Worked example · Membership and uniqueness

In a basis, let $\vec{u} = (1;0;1)$, $\vec{v} = (0;1;1)$ and $\overrightarrow{AM} = (2;3;5)$. We seek λ, μ such that $\overrightarrow{AM} = \lambda\vec{u} + \mu\vec{v}$. The first two coordinates require $\lambda = 2$, $\mu = 3$ and the third satisfies $5 = 2 + 3$. The point M lies in the plane through the point A with directions $\vec{u}, \vec{v}$. For $(2;3;4)$, the last equality fails: the point does not lie in it.

A4 · Partial guidance Are vectors $\vec{u} = (1;1;0)$ and $\vec{v} = (0;1;1)$ collinear? For which value of m is the vector $(2;m;3)$ a combination of $\vec{u}$ and $\vec{v}$?

A5 · Correct a mistake In an arbitrary basis, a pupil calculates the norm of the vector with coordinates $(1;1;0)$ as $\overline{2}$. Explain the missing assumption. Give an example with two unit vectors at an angle of 60 °.

10.4 From the solid to the calculation

First use: S17.

Worked example · Midpoints and parallelism

If I and J are the midpoints of [OA] and [OB], $\overrightarrow{IJ} = \overrightarrow{IO} + \overrightarrow{OJ} = -\frac{1}{2}\overrightarrow{OA} + \frac{1}{2}\overrightarrow{OB} = \frac{1}{2}\overrightarrow{AB}$. The lines (IJ) and (AB) therefore have collinear directions. This calculation holds in any non-degenerate triangle.

A6 · Choice of strategy In the tetrahedron $OABC$, I is the midpoint of $[OA]$, J that of $[OB]$ and K that of $[OC]$. Prove that the planes (IJK) and (ABC) are parallel, then that O does not belong to (IJK). Do not use a Cartesian equation.

A first step: Compare $\overline{IJ}$ to $\overrightarrow{AB}$ and $\overrightarrow{IK}$ to $\overrightarrow{AC}$. For membership, decompose in $(\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC})$.

One minute, spoken aloud. Describe the tetrahedron and the three midpoints without showing the figure. Your partner must be able to place them. Specify the segments rather than "the point on the left".

Delayed review · S51 then S70. The same plane will be described by a normal vector, then an equation. The vector definition remains the way to check the calculation.

Investigation

Three midpoints in a cube - 40 min.

Consider the cube ABCDEFGH shown above. Let $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are the respective midpoints of $[AB]$, $[AD]$ and $[AE]$. Let O the centre of the cube.

1. Express $\overline{IJ}$ and $\overrightarrow{IK}$ in the basis $(\vec{\imath}, \vec{\jmath}, \vec{k})$.

2. Prove that the points I, J and K are not collinear.

3. Determine whether O lies in the plane (IJK). Justify the answer by checking whether $\overrightarrow{IO}$ is a linear combination of $\overline{IJ}$ and $\overrightarrow{IK}$.

4. Why does reading the figure alone not allow us to prove either membership or non-membership of O in plane (IJK) ?

Progressive exercises

E1 · Direct application. In a basis of space, the vectors $\vec{u}$ and $\vec{v}$ have resp. coords $(1;2;-1)$ and $(3;6;-3)$. Are they collinear? Justify.

E2 · Application and justification. In an orthonormal coordinate system of space, consider $A(0;1;2)$ and $B(4;-1;0)$. Determine the coordinates of the midpoint of $[AB]$, then calculate the distance AB.

E3 · Method to organise. In a basis of space, consider $\vec{u} = (1;0;1)$, $\vec{v} = (0;1;1)$ and $\vec{w} = (2;3;5)$. Express $\vec{w}$ as a linear combination of $\vec{u}$ and $\vec{v}$, then deduce that these three vectors are coplanar.

E4 · Choice of strategy. In a basis of space, the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ have coordinates $(1;0;1)$, $(0;1;1)$ and $(1;1;0)$. Is the triple $(\vec{u},\vec{v},\vec{w})$ a basis of space? Justify.

E5 · Synthesis or proof. In a coordinate system of space, consider $A(1;0;2)$, $B(3;-1;4)$ and $C(5;-2;6)$. Determine whether C lies on the line (AB). Justify by vector calculation.

Work from the textbook or a past paper

Barbazo, p. 328, exercise 21. In class after S15: do the three questions. For each intersection, start by identifying each of the planes from the vertices of the cube and the given midpoints. Distinguish two planes that intersect, coincide or are strictly parallel.

Homework

DM10 - Points of a tetrahedron - 50 min.

Take a solid tetrahedron $ABCD$. The points I, J and K are the respective midpoints of $[AB]$, $[AC]$ and $[AD]$.

1. Express $\overline{IJ}$ in terms of $\overrightarrow{BC}$ and $\overrightarrow{IK}$ in terms of $\overrightarrow{BD}$.

2. Deduce that the planes (IJK) and (BCD) are parallel.

3. Define the point G by $\overrightarrow{AG} = \frac{1}{4}(\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD})$. Does point G lie in the plane (IJK) ? Justify.

4. For this question only, assume that, in an orthonormal coordinate system, $A(0;0;0)$, $B(4;0;0)$, $C(0;4;0)$ and $D(0;0;4)$. Calculate the lengths IJ, IK and JK.

MCQ A · consolidation (after S6)

After S6.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. In a basis, $(2;4;-2)$ and $(1;2;-1)$ are:

A. orthogonal B. collinear

C. non-coplanar D. equal

2. Three vectors $\vec{u}, \vec{v}, \vec{w}$, with $\vec{u}, \vec{v}$ non-collinear, are coplanar if:

A. $\vec{w} = \alpha\vec{u} + \beta\vec{v}$

B. $\vec{w} = 0$ only

C. their norms are equal

D. they have three coordinates

3. Two lines in space with non-collinear direction vectors are necessarily:

A. intersecting

B. non-coplanar

C. either intersecting or non-coplanar

D. parallel

4. In a basis, the expression of a vector is:

A. always impossible

B. unique when the three vectors form a basis

C. independent of the basis

D. only for orthonormal coordinate systems

5. An edge hidden in a perspective drawing must be:

A. erased

B. confused with a diagonal

C. shown with a different line style

D. declared not to exist

MCQ B · geometry review (after S17)

After S17.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. To show that $M \in (ABC)$, for A, B, C not collinear, we can:

A. compare AM to AB

B. write $\overrightarrow{AM} = s\overrightarrow{AB} + t\overrightarrow{AC}$

C. compute an area only

D. look at the figure

7. If two planes have the same direction, they are:

A. perpendicular

B. parallel or coincident

C. necessarily coincident

D. intersecting

8. In an orthonormal coordinate system, the midpoint of $A(0;1;2)$ and $B(2;3;4)$ is:

A. $(1;2;3)$

B. $(2;4;6)$

C. $(1;1;2)$

D. $(0;2;3)$

9. Two lines whose projections cross on the page:

A. intersect

B. are parallel

C. may be non-coplanar

D. coincide

10. For a triple to be a basis of space, it is enough to check:

A. non-coplanarity

B. orthogonality

C. equal norms

D. the initial point of the vectors

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. Two non-parallel lines in space intersect.

2. The zero vector is collinear with every vector.

3. A perspective drawing can serve as a proof of coplanarity.

Review of prior work

I can choose a basis, compute a linear combination and distinguish relative positions. **To watch out for:** mistaking an appearance of the figure for a property of space; forgetting a coordinate equation.

MCQ solutions

1. **B.** The first vector is twice the second. Moreover, their scalar product is not available in an arbitrary basis; collinearity does not depend on a metric.
2. **A.** With the first two non-collinear vectors, linear combinations characterise their vector plane. The zero vector is a particular case, not the only one.
3. **C.** Non-collinear direction vectors rule out parallel and coincident lines. The intersection remains to be established: the lines may also be non-coplanar.
4. **B.** A basis guarantees the existence and uniqueness of the decomposition. The coefficients depend on the choice of basis, which does not need to be orthonormal.
5. **C.** The dashed line keeps the edge and shows that it is hidden. Erasing the edge would remove information about the solid.
6. **B.** Non-collinear AB, AC span the direction of the plane. A relation $AM = sAB + tAC$ therefore proves membership.
7. **B.** Two planes with the same direction are parallel in the broad sense. A point must be tested to distinguish coincident planes from strictly parallel ones.
8. **A.** We average each pair of coordinates: $(1;2;3)$. This midpoint formula does not even require orthonormality.
9. **C.** Perspective is a projection that loses depth. A drawn intersection does not prove an intersection of the lines in space.
10. **A.** Three non-coplanar vectors form a basis. Orthogonality and norm 1 are not required.

True/false. False: the lines (AB) and (CG) of a cube are non-coplanar. True: $\vec{0} = 0\vec{u}$. False: a projection can create apparent intersections; a calculation or geometric proof is necessary.

Solutions to the applications and guidance

Application 1. $\overrightarrow{JK} = 2\vec{u} - \vec{v} - \frac{1}{2}\vec{w}$; $\overrightarrow{IK} = \frac{3}{2}\vec{u} - \frac{3}{2}\vec{v}$.

Application 2. The first two drawings place the lines in a common plane; the third represents opposite edges of a non-degenerate tetrahedron. In a cube, (AB) and (CG) are not coplanar: the only plane containing A, B, C is that of the face $(ABCD)$, which does not contain G.

Application 3. (OA) and (BC) are neither parallel nor intersecting. Add "coplanar". Two distinct lines in the same plane that are not parallel intersect.

Application 4. The first coordinate forces any ratio between $\vec{v}$ and $\vec{u}$ to be 0, impossible since $\vec{v} \neq 0$. The combination is $(\lambda; \lambda + \mu; \mu) : \lambda = 2$, $\mu = 3$, so $m = 5$.

Application 5. An orthonormal coordinate system is needed. With two unit vectors at an angle of 60 °, the square of the norm of their sum is $1 + 1 + 2\cos 60° = 3$; the norm is $\sqrt{3}$. The first time round, we can also see this length as a diagonal of a rhombus.

Application 6. $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{AB}$ and $\overrightarrow{IK} = \frac{1}{2}\overrightarrow{AC}$: two non-collinear directions of the planes are parallel, so the planes are parallel. If $O \in (IJK)$, then $\overrightarrow{IO} = \lambda\overrightarrow{IJ} + \mu\overrightarrow{IK}$. The coefficients of $\overrightarrow{OB}$ and $\overrightarrow{OC}$ require $\lambda = \mu = 0$, while that of $\overrightarrow{OA}$ would be $-\frac{1}{2} = 0$, contradiction.

Spatial investigation

Have students handle four non-coplanar points, then draw. In the solution, distinguish what is a convention of representation and what is proved. To help, provide the three Chasles decompositions; to go further, replace the midpoints with the same ratio $t \in]0; 1[$ and recover the parallelism.

Teaching notes and solutions

Guidance for the teacher

Written test 1 comes before S12 and S15: set vector geometry in a given basis, without require new coordinate techniques. Use a physical cube, then an oblique projection; mark the hidden edges. Two non-parallel lines in space are not necessarily intersecting. The norm via coordinates requires an orthonormal coordinate system. The formal parametric representations and equations of planes will be dealt with later.

Solutions to the activity

1. By Chasles' relation, $\overrightarrow{IJ} = \overrightarrow{AJ} - \overrightarrow{AI} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{\jmath}$ and $\overrightarrow{IK} = -\frac{1}{2}\vec{\imath} + \frac{1}{2}\vec{k}$.

2. If $\overrightarrow{IK} = t\overrightarrow{IJ}$, the coefficient of $\vec{k}$ would be $\frac{1}{2} = 0$, impossible in a basis. These two vectors are therefore non-collinear; I, J, K define a plane.

3. The centre of the cube satisfies $\overrightarrow{AO} = \frac{1}{2}(\vec{\imath} + \vec{\jmath} + \vec{k})$, therefore $\overrightarrow{IO} = \frac{1}{2}\vec{\jmath} + \frac{1}{2}\vec{k}$. If $O \in (IJK)$, there is s, t such that $\overrightarrow{IO} = s\overrightarrow{IJ} + t\overrightarrow{IK}$. Comparing the coefficients in the basis,

$$0 = -\frac{s+t}{2}, \qquad \frac{1}{2} = \frac{s}{2}, \qquad \frac{1}{2} = \frac{t}{2}.$$

The last two require $s = t = 1$, whereas the first would require $s + t = 0$. Contradiction : $O \notin (IJK)$.

4. Oblique (cavalier) perspective is a projection onto a sheet: distinct points may have the same image. It helps to conjecture, but an apparent overlap does not prove membership of a plane.

E1. The three coordinates of $\vec{v}$ are three times those of $\vec{u}$: $\vec{v} = 3\vec{u}$. The two vectors are therefore collinear.

E2. The midpoint has coordinates $(\frac{0+4}{2}, \frac{1-1}{2}, \frac{2+0}{2}) = (2, 0, 1)$. The difference of coordinates gives $\overrightarrow{AB} = (4, -2, -2)$; in the orthonormal coordinate system, $AB = \sqrt{16+4+4} = 2\sqrt{6}$.

E3. We calculate $2\vec{u} + 3\vec{v} = (2, 0, 2) + (0, 3, 3) = (2, 3, 5) = \vec{w}$. Thus $\vec{w}$ belongs to the vector plane spanned by the first two vectors: the three are coplanar.

E4. The first two vectors are non-collinear. If the third were a linear combination of them $\lambda(1, 0, 1) + \mu(0, 1, 1)$, the first two coordinates would require $\lambda = \mu = 1$, but the third would equal $2 \neq 0$. The three vectors are non-coplanar; they therefore form a basis of space.

E5. $\overrightarrow{AB} = (2, -1, 2)$ and $\overrightarrow{AC} = (4, -2, 4) = 2\overrightarrow{AB}$. Since $A \neq B$, the line (AB) is defined and this collinearity proves $C \in (AB)$.

Solutions to homework DM10

1. By Chasles' relation and the definitions of the midpoints,

$$\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{AC} - \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\overrightarrow{BC}, \qquad \overrightarrow{IK} = \frac{1}{2}\overrightarrow{AD} - \frac{1}{2}\overrightarrow{AB} = \frac{1}{2}\overrightarrow{BD}.$$

2. The points B, C, D are not collinear, because the tetrahedron is not flat. The vectors $\overrightarrow{BC}, \overrightarrow{BD}$ are therefore non-collinear, as are $\overrightarrow{IJ}, \overrightarrow{IK}$. The planes have the same direction. They are distinct: if $I \in (BCD)$, the line $(BI) = (BA)$ would be contained in this plane, which would place A, contradiction. They are therefore strictly parallel.

3. In the basis $(\overrightarrow{AB}, \overrightarrow{AC}, \overrightarrow{AD})$, we have $\overrightarrow{IG} = (-\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$, $\overrightarrow{IJ} = (-\frac{1}{2}, \frac{1}{2}, 0)$ and $\overrightarrow{IK} = (-\frac{1}{2}, 0, \frac{1}{2})$. A relation $\overrightarrow{IG} = s\overrightarrow{IJ} + t\overrightarrow{IK}$ forces $s = t = \frac{1}{2}$ by the last two coordinates; the first would then be $-\frac{1}{2}$, and not $-\frac{1}{4}$. Hence $G \notin (IJK)$.

4. In this particular case, $I(2,0,0)$, $J(0,2,0)$ and $K(0,0,2)$. We find $IJ = \sqrt{(-2)^2 + 2^2} = 2\sqrt{2}$, and also $IK = JK = 2\overline{2}$. The triangle is equilateral. This metric conclusion does not hold for every non-degenerate tetrahedron: it uses the coordinates of the particular case.

Solutions to the textbook or past-paper work

In this exercise, I is the midpoint of $[AB]$ and J that of $[CD]$.

1. The planes (AIE) and (BIG) both contain (AB). They are distinct: G does not belong to the plane of the face $ABFE$. Their intersection is therefore exactly the line (AB).

2. The points A, D, I are three non-collinear points of the face $ABCD$; the same holds for B, J, C. The two planes are therefore coincident with (ABC). The intersection is the whole plane, not a line.

3. The plane (HEF) is that of the top face; (BJC) is that of the bottom face. These planes are strictly parallel: their intersection is empty.